

P.V. Jayavardhana Sai

GenAI / LLM Systems Engineer — RAG — AI Agents — Python

LinkedIn

GitHub

EDUCATION

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LeetCode

- ICFAI University (IFHE)** Hyderabad, India
 - B.Tech - Computer Science Engineering — CGPA: 8.6* 2022 – 2026
 - Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases*

SKILLS SUMMARY

- Programming Languages:** Python, Java, C, C++ JavaScript, HTML5, CSS3, SQL
- ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Transformers, Librosa
- Full-Stack & Tools:** Flask, Docker, Git/GitHub, Jupyter, MySQL
- Core Expertise:** LLM Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Deep Learning, NLP, Computer Vision
- Soft & Leadership Skills:** Team Leadership, Agile Project Management, Stakeholder Communication, Problem Solving

EXPERIENCE

- INCOIS (Indian national centre for ocean information services)** Hyderabad, India
 - Research intern – Machine learning* January 2026 – Present
 - ML-Based Ocean Data Reconstruction:** Developed machine learning models (DINCAE) for spatial gap filling in tropical Indian Ocean datasets. Designed data pipelines for preprocessing, training, and evaluation of geospatial models. Analyzed model performance and failure cases to improve reconstruction accuracy and robustness.
- Wipro Limited** Hyderabad, India
 - Student Intern – Project Management* June 2024 – July 2024
 - Project Management & Stakeholder Coordination:** Collaborated with team of 5 members, managed timelines, and delivered comprehensive project reports to mid-level leadership
 - Recognized Contribution:** Awarded official Project Completion Certificate by Wipro View Certificate

PROJECTS

- Cognitive Resonance AI – Emotion-Aware Contextual AI Framework:** Architected a novel deep learning system integrating cognitive modeling with real-time emotional uncertainty quantification and context adaptation for human-centric AI (Manuscript under review, 2024–Present)
- Automated Student Section Allocation Platform:** Engineered full-stack solution (Python + React + MySQL) automating allocation for 1500+ students with rule-based optimization engine – eliminated 100% manual intervention and achieved 99.9% accuracy (Jun 2023–Aug 2024)
- Real-Time Speech Processing Interface:** Built end-to-end voice pipeline with noise-robust feature extraction (Librosa) and TTS synthesis (pyttsx3), boosting accessibility in low-resource environments (May–Aug 2023)
- Library Management System:** Designed scalable SQL database with automated alerts, fine calculation, and user dashboard (Mar–May 2023)
- Workshop AI Assistant – RAG (LLM System):** Designed and deployed a Retrieval-Augmented Generation (RAG) system over private vehicle service data using Ollama and Python. Built multi-step pipelines for query processing, context retrieval, and response generation. Implemented prompt orchestration and structured outputs to improve reliability. Identified and mitigated LLM failure modes such as hallucination, irrelevant retrieval, and context drift. Added logging and debugging layers to trace pipeline behavior and improve response quality.
- Portfolio & Source Code:** More projects accessible at github.com

CERTIFICATIONS

- Data Processing & Visualization:** NASSCOM FutureSkills
- Exploratory Data Analysis:** NASSCOM
- Python for Data Science & Machine Learning:** Kaggle
- Google Cybersecurity Professional Certificate:** Coursera
- Python Programming:** HackerRank – 5 Gold Badge
- Problem Solving (Intermediate):** HackerRank – 6 Gold Badge
- AWS Academy Cloud Foundations:** Amazon Web Services

ACHIEVEMENTS & LEADERSHIP

- CSIRO Image2Biomass Global Challenge 2025:** Achieved Public **Leaderboard Score of 0.60** (Top percentile) using DINOv2 + Multi-Scale Vision Transformer ensemble
- LeetCode :** Solved 250+ problems — Active participant
- ICFAI Tech Merit Scholarship:** Awarded 3,50,000 scholarship for ranking in Top 15% of batch
- Vice-President – LITGALOR English Literature Club:** Leading 20+ member team, organizing literary fests and workshops (2023–2025)
- Smart India Hackathon 2024 –:** Team Leader of 6-member team representing ICFAI University
- ARC - AGI2:** Competed in ARC 2025 AGI 2, developing solutions for abstract reasoning tasks